Quiz — 1.2.2 Applications Generation

Name: ___ Date: _ ***Mark:*** ___ / 25

OCR H446 · Component 01 · 1.2.2

Section A — Multiple choice (8 marks)

1. Which of the following is an example of **utility software**? A. Web browser B. Word processor C. Disk defragmenter D. Computer game Your answer: ☐

2. A key feature of **open source** software is that: A. The source code is hidden and cannot be modified B. The source code is available and may be modified C. It always requires a paid licence D. It can only be supported by the original vendor Your answer: ☐

3. A **compiler** differs from an **interpreter** because it: A. Translates and executes one line at a time B. Produces a standalone executable, translating all at once before running C. Requires the source code to be present every time the program runs D. Stops at the first error it finds Your answer: ☐

4. An **assembler** translates: A. High-level code into machine code all at once B. Assembly language into machine code (≈ one-to-one) C. Machine code into high-level code D. Bytecode into source code Your answer: ☐

5. In which stage of compilation are **syntax errors** reported? A. Lexical analysis B. Syntax analysis (parsing) C. Code generation D. Code optimisation Your answer: ☐

6. During **lexical analysis** the source code is: A. Checked against the grammar to build a parse tree B. Converted into tokens with whitespace/comments removed C. Turned into final machine code D. Refined to run faster Your answer: ☐

7. What is the purpose of a **loader**? A. To combine modules and libraries into one executable B. To copy the executable from storage into RAM ready to run C. To translate one line of source at a time D. To remove dead code from the program Your answer: ☐

8. A **dynamic** link (e.g. a .dll) differs from a static link because it: A. Copies all library code into the executable, making it larger B. Is linked at run time and can be shared/updated separately C. Removes the need for any libraries D. Can only be used with interpreted languages Your answer: ☐

Section B — Short answer (12 marks)

9. State **one** difference between application software and utility software, giving an example of each. **[2]**

10. The four stages of compilation are listed below in the wrong order. Write them out **in the correct order**. **[2]**

Code generation · Syntax analysis · Code optimisation · Lexical analysis

1. _____ **2.** _____

2. _____ **4.** _____

11. Explain **two** advantages of using an interpreter rather than a compiler when developing and debugging a program. **[2]**

12. Explain what a **library** is and give **two** reasons a programmer might use one. **[3]**

13. Describe the role of a **linker** and explain the difference between **static** and **dynamic** linking. **[3]**

Section C — Exam-style question (5 marks)

14. A developer has finished writing a program and wants to distribute it to customers who must not be able to see or change the source code, and who should not need any extra software to run it. Explain why a **compiler** is more appropriate than an **interpreter** for this purpose. **[5]**